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## **EE/EX-7002-CBGS**

### **B.E. VII Semester**

Examination, June 2020

## **Choice Based Grading System (CBGS)**

### **Electric Drives**

**Time : Three Hours**

**Maximum Marks : 70**

- Note:** i) Attempt any five questions.  
ii) All questions carry equal marks.

1. a) Explain the operation of a single phase semi-controlled converter fed separately excited D.C motor drive.  
b) A 230V, 750rpm 25A dc series motor is driving at rated condition a load whose torque is proportional to speed squared. The combined resistance of armature and field is  $1\Omega$  calculate the motor terminal voltage and current for a speed of 400rpm. State the assumption made for solving this problem.
2. a) Derive an expression for the average output voltage of a  $3\phi$  semicontrolled converter.  
b) The speed of a separately excited dc motor is controlled by means of a  $3\phi$  semi converter from a  $3\phi$ , 415V, 50Hz supply the motor constant are inductance 10mH resistance  $0.9\Omega$  and armature constant  $1.5\text{V/rad/sec (Nm/A)}$ . Calculate the speed of this motor at a torque of 50Nm when the converter is fired at  $40^\circ$ . Neglect losses converter.

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3. Explain with circuit diagram how regenerative braking is applied to
  - a) Separately excited dc motor and
  - b) Series dc motor
4.
  - a) Explain multiquadrant operation of a separately excited DC motor drive fed from a dual converter in detail.
  - b) A 220V, 1500rpm, 50A separately excited DC motor with armature resistance of  $0.5\Omega$  is fed from a circulating current dual converter with  $3\phi$  ac source voltage of 165V (line). Determine converter firing angles for the following operating points.
    - i) Motoring operation at rated motor torque and 1000 rpm.
    - ii) Braking operation at rated motor torque and  $-1000$  rpm.
5.
  - a) Compare the operation of VSI and CSI fed Induction motor drive.
  - b) Discuss variable frequency control of IM drive. Draw the relevant speed torque characteristics and derive the mathematical expression showing the relationship of max torque and operating frequency.
6.
  - a) Explain the principles of operation of VSI fed induction motor.
  - b) Show that a variable frequency induction motor drive develops at all frequencies the same torque for a given slip-speed when operating at constant flux.
7.
  - a) How do you explain the operation of an induction motor speed control using rotor resistance variation?
  - b) Explain the operation of an induction motor speed control using a chopper control.
8.
  - a) Explain the conventional scherbius system with that of a solid state scherbius drive.
  - b) Write short note:  
Separate and set control of synchronous motor.

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